

3. Use Case Name:

Create Supplier Order

Textual Description:

The system creates a supplier order when a product's inventory falls below the minimum threshold. The process includes finding the best available supplier, calculating the missing quantity, and sending the order request to the supplier system. This use case ensures timely replenishment of low stock items and maintains adequate inventory levels.

List of Actors:

- **Primary:** System (Automatic) OR Store Manager (Manual)
- **Secondary:** Supplier System (receives order)

Pre-conditions:

- Product inventory is below minimum threshold (for automatic)
- OR Store Manager initiates manual order (for manual)
- Suppliers exist in database for the product

Post-conditions:

- SupplierOrder record is created in database
- Order is sent to Supplier System
- Order status = "Sent" or "Pending"
- Confirmation is received from Supplier System

Main Success Scenario:

1. Check Minimum Stock detects low inventory (or Manager initiates manually)
2. System calls "Find Best Supplier"
3. System retrieves selected/best supplier
4. System calculates missing quantity
5. System creates SupplierOrder object with:
 - Order ID
 - Supplier ID
 - Product ID
 - Quantity needed
 - Order Date
 - Status = "PENDING"
6. System saves order to database
7. System calls "Send Order Request"
8. System receives confirmation from Supplier System
9. System updates order status = "SENT"
10. Use case ends successfully

Alternatives/Extensions:

- **3a: Manual Supplier Selection**
 - At step 2: Store Manager manually selects supplier
 - Bypasses "Find Best Supplier"
 - Continues from step 3
- **3b: Order Creation Failure**
 - At step 7: Supplier System rejects order
 - System updates order status = "FAILED"
 - System generates error alert
 - Manual retry option is provided
- **3c: Partial Order**
 - At step 4: If supplier has limited quantity
 - System creates order for available amount
 - Remaining quantity noted for future order

4. Use Case Name:

Find Best Supplier

Textual Description:

The system identifies the optimal supplier for a product when creating a supplier order. The selection is based on multiple criteria including supplier rating, lead time, availability, and historical performance. This ensures the most reliable and cost effective sourcing of products to maintain inventory levels.

List of Actors:

- Primary: System (Automatic)
- Secondary: Database (supplier data provider)

Pre-conditions:

- Product needs ordering
- At least one supplier exists for the product
- Supplier ratings/metrics are available

Post-conditions:

- Best supplier is identified
- Supplier information is retrieved
- Supplier is ready for order creation

Main Success Scenario:

1. System initiates supplier search for product
2. System queries database for all suppliers of product
3. System retrieves supplier list from ProductSupplier table
4. For each supplier, system retrieves:
 - Supplier ID
 - Supplier Name
 - Supplier Rating (0-100)
 - Lead Time (days)
 - Availability Status
 - Historical Performance Score
5. System calculates combined score for each supplier
6. System identifies supplier with highest score
7. System retrieves full details of best supplier
8. System returns best supplier to calling use case
9. Use case ends successfully

Alternatives/Extensions:

- 4a: No Suppliers Available
 - At step 3: If supplier list is empty
 - System generates error
 - Manual supplier selection required
- 4b: Supplier Unavailable
 - At step 6: If best supplier is unavailable
 - System selects next best supplier
 - Continues from step 7
- 4c: Manual Override
 - Store Manager can manually select supplier
 - Bypasses automatic selection
 - Confirms override reason

5. Use Case Name:

Send Order Request

Textual Description:

The system sends the prepared supplier order to the external Supplier System. The order request includes all necessary details about the product, quantity. The system waits for and processes the confirmation response from the Supplier System, updating the order status accordingly.

List of Actors:

- **Primary:** System (Order sender)
- **Secondary:** Supplier System (Order receiver)

Pre-conditions:

- SupplierOrder has been created
- Order details are complete and validated
- Connection to Supplier System is available
- Order has status = "PENDING"

Post-conditions:

- Order is transmitted to Supplier System
- Confirmation response is received
- Order status is updated in database
- Delivery date is recorded

Main Success Scenario:

1. System prepares order for transmission:
 - Formats order data
 - Includes: supplier_id, product_id, quantity, order_date
2. System establishes connection to Supplier System
3. The system sends requests to the Supplier System.
4. Supplier System validates order
5. Supplier System processes order
6. Supplier System sends confirmation response with:
 - Confirmation Status = "SUCCESS"
 - Supplier Order ID
 - Expected Delivery Date
 - Estimated Cost
7. System receives and parses confirmation
8. System updates SupplierOrder record:
 - Status = "SENT"
 - Supplier Order ID (from response)
 - Expected Delivery Date
9. Use case ends successfully

Alternatives/Extensions:

- **5a: Supplier Rejects Order**
 - At step 6: Supplier System returns error
 - Status = "REJECTED"
 - System suggests alternative supplier

